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Generation of Immune Stimulating Peptides (ISPs) for Refractory and
Recurrent Pediatric Cancers

by

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A THESIS

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Abstract

Pediatric cancer ranks among the top causes of childhood mortality in developed nations, presenting persistent challenges despite the availability of conventional therapies like chemotherapy. Refractory and recurrent cancers, especially prevalent in the pediatric population, underscore the urgent need for innovative treatments. Immune stimulating peptides (ISPs) emerge as a promising avenue, targeting tumor-specific antigens – known as neoantigens – to trigger anti-tumor immune responses. The purpose of this study was to devise a bioinformatics pipeline capable of generating ISPs for diverse cancers. The pipeline utilizes various bioinformatics and machine learning tools to identify potential neoantigens and predict their ability to elicit an immune response. Tested on two genetically distinct ependymoma subclasses, the pipeline demonstrated its effectiveness by generating 20 and 24 predicted immunogenic peptides for each subclass, respectively. These results showcased the pipeline’s efficacy in generating ISPs tailored towards unique cancer cases. Moreover, marking a significant advancement towards developing a novel treatment for refractory pediatric ependymoma.

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Reflection

The process of conducting research and writing this honours thesis was an invaluable experience. It helped to develop and hone my skills in all facets of research, from planning and executing a study to working in a lab environment. As a bioinformatics student I have not taken any immunology classes whatsoever, so going into this project I knew next to nothing about how the immune system fought cancer, and the mechanisms involved. This was an exercise in parsing literature to gain knowledge on a given topic, a skill that I have been developing over the past 4 years. To me, this highlighted that the value of an undergraduate degree is not necessarily in the content you memorize but the skills you learn. For me particularly these skills are the ability to learn a topic through literature, lab skills, programming, and data management, all of which were tested over the course of this research project.

I plan to continue applying and developing these skills by conducting more research in the future. The field of cancer treatment fascinates me, and I believe that bioinformatics will play a pivotal role in the future of the field. I plan to keep expanding on my thesis work in the coming years and am excited to see where I can take it and where it can take me.

The best part of this experience was having freedom in conducting a project that has the potential to positively impact people's lives. It was truly a fulfilling experience. This freedom also

presented challenges in staying disciplined, especially with other courses and commitments. Although different from anything I am used to, the freedom of the experience was positive overall and gave me a taste of what my future in academia might look like.

Dedication

I dedicate this work to my parents for their unwavering belief in me, to my friends, Austin, Nico, Krina, Arthur, and Onyi for your support throughout my academic journey, and to my brother Ben for always being someone I could look up to.

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The Current Treatment Landscape of Pediatric Cancers

In 2021, childhood malignancies emerged as the second leading cause of mortality among individuals aged 1 to 14 in Canada, surpassed only by unintentional injuries [1]. This trend is not isolated to Canada alone; across developed nations, childhood cancer places highly in the leading causes of childhood mortality, ranked second in Europe in 2016 [2] and second in the USA in 2021 [3]. Despite the efficacy of conventional cancer treatments such as chemotherapy, a challenge is posed by refractory cancers – those displaying an inherent or adaptive resistance under selective pressure [4]. The consequence of this resistance is a heightened risk of cancer recurrence, presenting difficulty in subsequent treatment efforts.

In general, conventional cancer treatments like chemotherapy work to target rapidly dividing cells, a characteristic hallmark of malignancies. The inherent heterogeneity within cancer cell populations allows subsets to adopt a slow cycling phenotype, enabling the survival of these treatments [5]. These quiescent cancer cells can eventually reassume the proliferative phenotype, resulting in relapsed cancer cells possessing a heightened resistance to treatment [5]. Refractory and recurrent cancers present the greatest challenges to the current cancer treatment landscape and highlight the need for alternative treatment options.

Immune Stimulating Peptides (ISPs)

A promising new candidate in the field of cancer treatments are ISPs. ISPs are immunotherapies, meaning they aim to utilize the body's immune system to target and eliminate malignancies. Due to the mutational frequency seen in cancer cells, they will express unique antigens on their cell surface, not found elsewhere in the body; these unique antigens are known

as neoantigens. Neoantigens are generated when peptides distinct to the cancer are processed by the proteasome into shorter segments (~8-11 amino acids), which are then presented on the cell surface by major histocompatibility complexes (MHC). Since neoantigens are unique to the tumor and not found elsewhere in the body, they present themselves as tumor identifiers and a potential means of targeting via immunotherapies. ISPs function as an immune therapy by the introduction of a peptide sequence to the body corresponding to a neoantigen [6]. This is intended to enhance the patient's T-cell-mediated immune response, enabling it to identify and eliminate the tumor [6,7,8].

The immune system naturally fights cancer through a sequence of seven repeating steps, collectively termed the cancer-immunity cycle. 1) Neoantigens from the cancer cells are released and captured by dendritic cells (DCs). 2) These DCs present the captured antigens on class I and II MHC molecules to T cells, 3) triggering the activation of cytotoxic T cell (CTL) responses specifically targeting tumor neoantigens. 4) CTLs, once activated, proceed to migrate towards the tumor site. 5) They infiltrate the tumor and 6) engage in interactions involving their T cell receptors (TCR) and the target antigens bound to MHC-I on the surface of the target cell. 7) This interaction enables the T cells to eliminate the target cancer cells, releasing additional tumor-associated antigens and restarting the cycle (Figure 1) [9].

While this immune response is crucial in preventing and fighting cancer, it also imposes selective pressure on cancer cells, enabling them to adapt to evade various steps in the cancer-immunity cycle. For example, cancer cells can evade step 2 of the cancer-immunity cycle by inhibiting DC differentiation and maturation through the release of tumor-derived factors such as VEGF, IL-10, and TGF- β [10].

Thus, the premise of an ISP is to kickstart the cancer-immunity cycle by simulating the release of tumor-specific neoantigens that occurs in step 1. This primes a T-cell mediated immune response against cells expressing the tumor neoantigens on their surface. Notably, this addresses the major gaps in the current cancer treatment landscape by providing an alternative treatment for refractory cancers [11,12] and by reducing the risk of relapse through immune memory [13]. Thus, ISPs present themselves as promising candidates to be used both as an alternative to and in conjunction with conventional treatments.

Identifying High Potential Neoantigens

Neoantigens arise from non-synonymous somatic mutations resulting in unique peptides which are displayed on tumor cell surfaces. The characteristic high mutagenicity and rapid proliferation of cancer leads to a frequent occurrence of such mutations, resulting in numerous potential neoantigens [14]. Therefore, the key to designing effective ISPs lies in identifying those tumor-specific peptides with the highest potential for surface presentation.

Cellular proteins undergo proteasomal processing, generating shorter peptide sequences presented as antigens via interactions with MHC-I and MHC-II. MHC-I primarily presents short peptides (~8-11 amino acids) derived from endogenous proteins, including those produced through non-synonymous somatic mutations in cancer cells. Presentation by MHC-I is essential for the surveillance and response of CD8⁺ CTLs. Conversely, MHC-II presents longer peptides (~13-25 amino acids) derived from exogenous proteins, such as those from pathogens. This process is central to activating helper CD4⁺ T cells [15]. While MHC-II is more commonly associated with the presentation of exogenous antigens, it is important to note that in some circumstances, MHC-II presentation of endogenous antigens can also occur [16,17]. When designing ISPs, it is crucial

to prioritize mutant proteins that could produce many possible unique peptide sequences after proteasome and protease processing. Such proteins are more targetable because they generate a greater diversity of potential neoantigens, maximizing the number of MHC binders that could be derived from a single peptide sequence.

There are three primary types of non-synonymous somatic mutations from which neoantigens can arise, each presenting unique advantages and challenges in terms of targetability and immunogenicity [14]. These mutation types include:

- 1. Single-Nucleotide Variants (SNVs):** SNVs involve the substitution of a single nucleotide base. They are the most common type of somatic mutation in cancer and result in peptide sequences that differ by only one amino acid. Consequently, this limits potential neoantigen peptides to those containing the altered amino acid.
- 2. Insertions and Deletions (INDELS):** INDELS result from the insertion or deletion of nucleotide sequences. Although less common than SNVs, INDELS can create frameshift mutations, leading to the generation of a higher number of potential neoantigens, making them more targetable. In fact, INDELS generate up to three times as many predicted neoantigens as SNVs alone [18]. Additionally, INDELS exhibit enhanced immunogenicity, giving rise to nine times as many neoantigens with strong MHC binding affinities where the wild-type sequence was not predicted to bind strongly [18].
- 3. Gene Fusions:** Gene fusions occur when two independent genes combine to form a hybrid gene via a chromosomal rearrangement. Gene fusions are the rarest of the three mutation types. Also capable of resulting in frameshifts, gene fusions generate six times as many candidate neoantigens as SNVs and INDELS combined [19]. Furthermore,

gene fusions are more likely to give rise to highly immunogenic neoantigens compared to SNVs and INDELS [19].

When designing ISPs, it is also important to target an oncogenic driver that is prevalent across a significant proportion of the patient population. The choice of an oncogenic driver is strategic, as the tumor will be less likely to adapt lose the targeted neoantigen, given its essential role in maintaining the cancerous phenotype. Furthermore, by selecting a neoantigen that is common among many patients, the ISP can serve as a therapeutic approach with broader applicability, benefiting a larger patient population rather than being specific to only a few individuals.

The Impact of Sex and Gender on Immunotherapies

In the realm of biomedical research, considering the potential effects of sex and gender on the outcomes of immunotherapeutic studies is essential. While this study focuses on ISPs and their potential as a novel treatment for pediatric cancers, it is crucial to acknowledge that factors related to sex can influence the immune response and treatment outcomes. Research has indicated that there are often sex-based differences in immune function, with females tending to mount a more robust immune response than males [20]. Additionally, hormonal fluctuations and genetic variations due to sex can impact the efficacy of immunotherapies [20].

Furthermore, gender identity and sociocultural factors may play a role in the acceptance and adherence to novel treatment modalities. Understanding the diverse perspectives and experience of individuals across the gender spectrum is valuable in the context of clinical trials and the development on personalized treatment approaches.

Ependymoma

In the context of pediatric cancer and the exploration of ISPs as a potential treatment, ependymoma presents itself as a compelling case for such a therapeutic approach. Ependymoma is a malignancy affecting the ependymal cells of the central nervous system (CNS) [21]. These glial cells line the ventricles of the CNS and serve a crucial role in the production of cerebrospinal fluid, a substance that protects and nourishes the brain and spinal cord [22]. Ependymoma is the third most common intracranial malignancy in children and is characterized by a poor clinical outcome and unpredictable disease behaviour, particularly in pediatric cases [23,24]. Conventional treatment of ependymoma first involves surgical resection of the tumor followed by radiation therapy and/or chemotherapy [23]. However, ependymoma is typically refractory to conventional treatments and has a relapse rate of approximately 50%; treatment options also become more limited following recurrence [25]. Not only is ependymoma typically unresponsive to conventional treatments, but sometimes – depending on the exact location of the tumor – total resection of the tumor may be infeasible, adding to its refractory nature since surgery is the primary treatment [21].

Ependymoma occurs in three CNS regions: 1) the spine, 2) below the cerebellar tentorium in the posterior fossa, and 3) above the cerebellar tentorium in cerebral tissue. Respectively, ependymomas occurring in these regions are named spinal ependymoma (SP-EPN), posterior fossa ependymoma (PF-EPN), and supratentorial ependymoma (ST-EPN) [26]. Furthermore, ependymoma can be distinctly characterized into nine molecular subgroups based on methylation profiling [26]. Five of these nine subgroups are anaplastic, and two of these are characterized by distinct oncogenic driving mutations: ST-EPN-YAP1 and ST-EPN-RELA [26]. Moreover, these two supratentorial ependymomas predominantly affect the pediatric and adolescent population, with ST-EPN-RELA being characterized by a poor 10-year overall survival (OS) probability of

49% [26]. Both subclasses also have a poor 5-year progression free survival (OS) probability, with ST-EPN-YAP1 being 66%, and ST-EPN-RELA being 29% [26]. Progression free survival indicates the length of time during and after treatment during which a disease does not worsen or progress. So, in the case of ST-EPN-RELA, only 29% of individuals will not experience progression within the first five years of treatment. So, PFS indicates that both ependymoma subclasses have low rates of responsiveness to conventional treatments. This combined with the fact that they are characterized by distinct and known oncogenic driving mutations makes them suitable candidates to test a bioinformatics pipeline aimed at generating ISPs tailored to cancers exhibiting different oncogenic profiles.

Research Objectives

Objective 1: Develop ISP Discovery Platform

The primary objective of this study is to design a semi-automated discovery platform that can effectively predict ISPs for a given cancer type. At the heart of this discovery platform will be a bioinformatics pipeline with focus on ensuring accuracy in predicting peptides that exhibit strong binding affinity with MHC molecules. Importantly, the platform will maintain flexibility across cancer types, ensuring adaptability for the generation of novel therapeutic options for many cancers. This versatility is crucial in expanding the potential applicability of the platform, making it a valuable tool for addressing diverse cancer scenarios that may benefit from personalized ISPs.

Objective 2: Test the Discovery Platform on Ependymoma Subclasses

The second objective of this study will be the validation of ISPs generated for two distinct ependymoma subclasses. Assessing the ability of the platform to generate predicted peptides for

each of these subclasses will demonstrate the flexibility and broader applicability of the platform in generating ISP sequences. Specifically, this will highlight the platform's potential to contribute to the development of effective therapeutic strategies for otherwise refractory cancers. Together these objectives address the research question: Can a discovery platform be developed with the capability to generate ISPs for a variety of cancers?

Methods

Transcriptomic Data

The initial steps in our pipeline (Figure 2A) involved the identification of neoantigen sequences. We obtained transcriptomic (RNA-seq) data for each ependymoma subgroup from the Sequence Read Archive (SRA) [27]. In total, 5 samples were downloaded, this included 3 samples of patients with ST-EPN-RELA and 2 samples of patients with ST-EPN-YAP1. The samples varied in sex, age, and histological grade (Table 1). We chose RNA-seq data due to its ability to offer insights into the protein expression profile of tumors, thus encompassing the landscape of expressed antigens. The RNA-seq data, downloaded in fastq format, underwent quality assessment using FastQC [28]. FastQC generated a comprehensive summary report evaluating the quality and integrity of the sequencing data. This report included per-base sequence quality scores, per-sequence quality scores, sequence quality distribution, and other pertinent metrics. This quality assessment step was pivotal, as the subsequent stages of the pipeline relied on the accuracy of the RNA-seq data. Poor or inaccurate reads at this stage would compromise the effectiveness of the entire pipeline.

Neoantigen Identification

The oncogenic drivers for each ependymoma subclass are gene fusions. To determine the exact amino acid sequence of these fusions and ensure its consistency within subclass populations, fusion identification software was employed. First, we aligned the transcriptomic data to the hg38 reference genome using the STAR RNA-seq aligner [29]. Alignment to a reference genome is crucial to make sense of the raw RNA-seq data by mapping each read to a location in the genome and thus determine the identity of each of the transcripts present in the sequencing data. The STAR aligner accepts sequencing data in fastq format and outputs aligned reads and splice junctions. The output of splice junctions is vital to the identification of gene fusions. The STAR output was funneled directly into the fusion identification software Arriba [30], a bioinformatics software that takes alignment and splice junction data as an input and outputs a ranked list of gene fusion candidates in the transcriptomic data and their precise amino acid sequences.

MHC Binding Prediction

The amino acid sequences of the neoantigens were then fed into two machine learning algorithms known as NetMHCpan-4.1 and NetMHCIIpan-4.0 [31]. These two machine learning algorithms utilize artificial neural networks (ANNs) to predict the binding of input peptides to MHC molecules [31,32]. ANNs consist of layers of computational units called neurons. These neurons receive inputs, perform computations, and generate outputs. Typically, ANNs consist of an input layer for receiving raw data and executing initial computations, one or more hidden layers where intermediate computations occur, and an output layer for performing the final computation and producing a prediction. During training, the network learns from a set of training data through iterative processes. After each iteration, the network's predictions are evaluated against actual

results, and adjustments are made to the computations of each neuron. This iterative training refines the network's ability to make accurate predictions over time.

NetMHCpan-4.1 and NetMHCIIpan-4.0 comprise three layers each: an input layer that receives the peptide and MHC sequence, a single hidden layer, and an output layer generating binding predictions scores using two metrics: binding affinity (BA) and eluted ligand (EL) [31,32]. BA data refers to quantitative measurements of the strength of the interaction between a ligand and its receptor; EL data refers to experimental data obtained from techniques where a mixture of ligand molecules is passed through a column or matrix containing an immobilized receptor. Both ANNs were trained on BA and EL data. The training dataset for NetMHCpan-4.1 is composed of over 13 million data points, encompassing 250 unique MHC-I molecules, whereas the training set for NetMHCIIpan-4.0 is composed of over 4 million data points covering 116 unique MHC-II molecules [31].

To measure the performance of these models, researchers, use metrics known as sensitivity, specificity, and receiver operating characteristic area under the curve (ROC AUC). Sensitivity refers to the true positive rate and is defined as the ratio of true positives to the sum of true positives and false negatives. Thus, sensitivity measures the proportion of correctly identified positive cases. Specificity refers to the true negative rate and is defined as the ratio of true negatives to the sum of true negatives and false positives; it measures the proportion of correctly identified negative cases. ROC AUC measures the area under an ROC curve which plots the true positive rate (sensitivity) against the false positive rate; an ROC AUC of 0.5 would indicate that the model is making random predictions, where an ROC AUC of 1.0 would indicate perfect performance. In the case of NetMHCpan-4.1 the data on the measurement of model's performance has not yet been released, but NetMHCpan-4.0 (the version directly preceding NetMHCpan-4.1) showcased a 95%

sensitivity in predicting ligands ranked within the top 1% of peptides within the source proteins [32]. It also exhibited a 98.5% specificity when using 2% rank threshold [32]. NetMHCpan-4.0 also demonstrated an ROC AUC of above 0.975 when combining EL and BA data [32]. NetMHCIIpan-4.0, on the other hand, did not measure sensitivity or specificity, but showcased an ROC AUC of 0.836 indicating that the model has a good ability to accurately distinguish between true positives and false positives [33].

The fusion sequence for each of the ependymoma subclasses were fed into NetMHCpan-4.1 and NetMHCIIpan-4.0 and were predicted for known human MHC supertypes, which are groups of human MHC molecules with similar peptide binding features. This produced a large output file containing BA and EL prediction scores for different segments of the input peptide for each predicted supertype.

Output Processing and Epitope Ranking

To select the peptides that could be used to generate anti-tumor immune response, some data processing needed to be done. To achieve this, a python script was written to perform several data manipulations on the raw csv output of the machine learning algorithms. Firstly, the raw output was cleaned up into a more readable format and subset into each MHC supertype. This allowed for the selection of the top binding peptides for each supertype, which was important to generate a set of binding peptides with coverage over each supertype.

The machine learning algorithms predict the binding of short segments of the input peptide, as such some of the output peptides did not contain the fusion breakpoint. These were filtered out, as they would have no effect in targeting the desired neoantigen, since the fusion is precisely what is being targeted.

Finally, the top binding peptides were selected by taking into consideration both the EL and BA ranks. This was done by computing the average of the two and then selecting the best ranked average; a favourable ranked average would fall below 10, as ranks closer to 0 indicate better predicted binding by both metrics. This way both metrics were accounted for, meaning for a peptide to be chosen it had to have favourable binding scores for both metrics. This step of the bioinformatics pipeline accounts for the most selective part of the cancer immunity cycle which is the presentation of neoantigens on antigen presenting cells via interactions with MHC molecules.

Using this python script the top five binding epitopes for each MHC supertype were selected. These peptides were considered for further design in the subsequent steps of the pipeline.

Antigen Processing Prediction

Another crucial step in the immune response against cancer is the processing of tumor peptides by the proteasome into short neoantigen epitopes. Thus, the proteasomal cleavage sites for each of the generated binding peptides was predicted. This was done using another ANN known as NetChop-3.1 [34,35]. The neural network architecture of NetChop-3.1 consists of an input layer to receive an amino acid sequence, a single hidden layer, and an output layer that provides the predicted cleavage sites [34]. The model was trained on over 5000 MHC-I ligands from the SYFPEITHI database [34]. Since the totality of MHC ligands are produced via proteasomal processing, they provide themselves as a valid training set for a machine learning model to predict proteasomal products. NetChop displays a specificity of 0.48, a sensitivity of 0.81, and an ROC AUC of 0.85 [34].

This step of the pipeline allowed for us to take different design approaches to longer peptides to be used to stimulate the immune response. First, we aligned the top binding peptides

for each ependymoma subclass around the fusion breakpoint and made a consensus peptide. Second, we predicted the cleavage sites for each of the smaller peptides and were able to group them into slightly larger peptides that, in theory, would be processed into each of the smaller subunits. This was done by taking advantage of predicted proteasomal cleavage. To maximize coverage, we then linked all these cleavage group peptides with a lysine-lysine (KK) and well as an alanine-alanine-tyrosine (AAY). KK and AAY function as cleavable linkers meaning the subunits linked by each of these should be products of proteasomal cleavage within the cell. The idea is that these long peptides would go through a cascade of proteasomal cleavage where they are first processed into the cleavage group subunits which can then be further processed into the predicted binding epitopes.

Results

Predicted MHC Binding Epitopes

The binding of a total of 126 MHC-I epitopes and 646 MHC-II epitopes was predicted for the ST-EPN-RELA subclass. In addition, the binding of 140 MHC-I epitopes and 646 MHC-II epitopes was predicted for the ST-EPN-YAP1 subclass. The python script narrowed these numbers down to the top 5 binding epitopes for each MHC supertype that contained the desired fusion breakpoint. Of these, most of the MHC-I binding epitopes exhibited favourable BA and EL scores (ST-EPN-RELA mean = 16.5; ST-EPN-YAP1 mean = 16.7), while in contrast almost none of the predicted MHC-II binding epitopes exhibited scores within the favourable range (ST-EPN-RELA mean = 81.8; ST-EPN-YAP1 mean = 42.4). There was overlap between the top binding epitopes for each supertype. With this overlap considered, the total of predicted functionally binding epitopes was 20 MHC-I binding epitopes for ST-EPN-RELA (Figure 3A), 18 MHC-I binding

epitopes for ST-EPN-YAP1 (Figure 4A), and 6 MHC-II binding epitopes for EPN-YAP1. The MHC-I epitopes have total combined coverage over all 12 MHC-I supertypes for each subtype, while the MHC-II epitopes for EPN-YAP1 only cover 3 of the 34 MHC-II supertypes.

Predicted Cleavage Group Peptides

By taking advantage of proteasomal cleavage site prediction, the predicted functional epitopes were grouped into cleavage group peptides – which are peptides that would be processed into the functional epitopes by the proteasome. Note that this was done only for the MHC-I peptides to maximize MHC supertype coverage. With this the 20 ST-EPN-RELA MHC-I binding epitopes were reduced into 6 cleavage group peptides (Figure 3B), and the 18 ST-EPN-YAP1 MHC-I binding epitopes into 3 cleavage group peptides (Figure 4B).

Further Peptide Design

We employed two different strategies for creating longer peptides consisting of the cleavage groups peptides: 1) consensus peptides and 2) linked peptides. Consensus peptides of the cleavage group peptides were constructed by centering around the fusion breakpoint and extending on either side until the peptides were 20 residues long. The cleavage group peptides for each ependymoma subclass were also linked to one another using cleavable KK and AAY linkers. The result was a 77-residue KK-linked peptide and an 83-residue AAY-linked peptide, each consisting of the 6 EPN-RELA cleavage group peptides. Produced in addition was a 43-residue KK-linked peptide and a 45-residue AAY-linked peptide, each consisting of the 3 EPN-YAP1 cleavage group peptides.

The proteasomal cleavage sites for each of the consensus peptides and the linked peptides were predicted. The consensus peptides were predicted to be cleaved into only some of the cleavage group subunits. In contrast both the KK-linked and AAY-linked peptides were predicted to be processed into *all* cleavage group subunits.

Discussion

Evaluating the Discovery Platform

Overall, the platform performed well in generating testable predicted ISPs. In total it generated 21 different peptides consisting of 9 for EPN-RELA MHC-I (6 cleavage group; 1 consensus; and 2 linked), 6 for EPN-YAP1 MHC-I (3 cleavage group; 1 consensus; 2 linked), and 6 for MHC-II EPN-YAP1 (all functional epitopes). The platform seemed to favour MHC-I, with most predicted MHC-II BA and EL rank scores for *both* ependymoma subclasses being far outside the favourable range of <10 (EPN-RELA mean = 81.8; EPN-YAP1 mean = 42.4). This was somewhat of an expected outcome as most anti-cancer ISPs seem to favour the use of the MHC-I pathway over the MHC-II pathway [36].

This could be due to a greater allelic diversity in MHC-II across the human population than MHC-I, like what is observed in some avian species [37]. This would lend the MHC-I binding peptides to a better inherent coverage between alleles. This would explain why most MHC-I binding epitopes exhibited favourable predicted BA and EL scores *across many* different MHC supertypes. It would also explain why this was not the case for MHC-II binding epitopes, as only some MHC supertypes exhibited favourable binding scores. However, according to the IPD-IMGT/HLA database, the number of human MHC-I alleles (26,610) doubles that of human MHC-II alleles (11,398) [38], which does not confer with human MHC-II possessing a greater allelic

diversity. Though, a higher single nucleotide polymorphism / 10kbp rate can be observed in human MHC-II alleles than can be observed in human MHC-I alleles [39], suggesting that the allelic diversity of human MHC-II is indeed greater than that of human MHC-I despite having a lower number of total alleles among the population.

Due to the platform favouring the generation of MHC-I binding peptides, the majority (15/21) of the output peptides function through the MHC-I pathway. This is favoured as MHC-I is involved in two steps of the cancer immunity cycle, while MHC-II is involved in only one [9]. These steps are the presentation of the neoantigen to T-cells (which can occur via MHC-I or MHC-II) and presentation of the neoantigen on the surface of tumor cells (which can *only* occur via MHC-I). Thus, it is necessary for the predicted peptides to bind MHC-I as this is a requirement for the tumor to be targetable by the neoantigen. The generation of MHC-II binding peptides was still performed as both CD8⁺ and CD4⁺ T-cell responses have been shown to act in anticancer immunity [40,41]. Additionally, there is a growing body of research demonstrating the efficacy of MHC-II based ISPs [42] and the role of CD4⁺ T-cell mediated cancer immunity specifically in conjunction with a CD8⁺ response [41,43]. Thus, it was important to have the platform be able to generate both MHC-I and MHC-II binding peptides to keep its applicability as wide as possible, especially in this growing body of research.

Project Limitations

One limitation of this project lies in the rarity of ependymoma as a disease. Despite being the third most common intracranial neoplasm [23], it is still quite rare, affecting only 200 children per year in the United States [44]. Due to this, the dataset comprising supratentorial ependymoma patient transcriptomic data is notably limited in sample size. Furthermore, there is a paucity in the

literature detailing the genetic make-up of the different ependymoma subclasses. These two factors led to our reliance on the fusion identification software, Arriba [30], to analyze a limited number of transcriptomic samples. This solely informed the subsequent steps in generating the ISPs. Although the identified fusions were consistent across samples, there may be rarer fusions involving the same genes but occurring at different breakpoints, which could also contribute to the oncogenesis of the two ependymoma subtypes. However, the peptides generated in this study would fail to initiate an immune response against neoantigens of these variants.

Another consequence of the limited literature classifying the genetic make-up of ependymoma subclasses meant that not all ependymoma subclasses could be targeted. Initially a posterior-fossa subclass (PF-EPN-A) was also meant to be targeted as it is the most common subclass in children and is also characterized by a poor survival rate [26]. Any consistent oncogenes for PF-EPN-A have yet to be identified, this was partially validated by our preliminary analysis in which Arriba identified 0 consistent gene fusions between PF-EPN-A transcriptomic samples. However, this does not discount the possibility of the existence of a consistent SNV or INDEL in this subclass. So, the use of ISPs to target PF-EPN-A could still be viable.

Another limitation of the study is that it is solely predictive. Throughout the pipeline various bioinformatics tools and machine learning algorithms are utilized to predict sequence identity, MHC binding affinity, and proteasomal cleavage sites. Although, these tools are accurate there is a margin for error in their predictive capabilities. The fusion identification algorithm Arriba [30] ranks among the best, showcasing the highest AUC values out of 24 different fusion identification algorithms [45]. However, it was still unable to identify *every* known fusion in

experimentally validated breast cancer lines [45]. This means that Arriba could have missed some gene fusions present in the ependymoma transcriptomic patient data.

The three machine learning algorithms used are also not guaranteed to be accurate. Recall that ROC AUC is a measure of the ability of a classification model to distinguish between classes, with a value of 1.0 being perfect discrimination and 0.5 being completely random. The two machine learning algorithms used to predict epitope binding to MHC complexes, NetMHCpan-4.0 and NetMHCIIpan-4.0, showcased ROC AUC values of 0.975 and 0.836 respectively [32,33]. And NetChop-3.1, the machine learning algorithm used to predict proteasomal cleavage sites, demonstrated an ROC AUC of 0.85 [34]. Although, each model outperforms random classification, there remains a possibility of misclassification, this is particularly true for NetMHCIIpan-4.0 and NetChop-3.1.

NetChop-3.1 specifically displayed a very low specificity of 0.48 [34]. This indicated that the model correctly identified 48% of the true negatives out of all the actual negatives. This means that when using the model false negatives are likely overrepresented in the models' predictions. This low specificity of NetChop-3.1 indicates a major fault in its predictive capability. However, all other specificity and sensitivity measurements for the three machine learning models fall within the acceptable range. These metrics indicate that although all these models are accurate in making predictions, their correctness is not guaranteed. This highlights the need for experimental validation of the output peptides.

Future Directions

The above limitations leave room for future research to improve both the performance of the model as well as add to the landscape of ependymoma literature. First, more research can be

done into the molecular classification and genetic make-up of different ependymoma subclasses, particularly PF-EPN-A. This is because PF-EPN-A remains the most aggressive pediatric ependymoma subclass for which a consistent oncogene has yet to be identified. If this could be accomplished then ISPs could greatly improve the entirety of the ependymoma treatment landscape, rather than just that of supratentorial ependymoma.

In addition to this, a study should be done to procure a greater number of ependymoma samples of each subclass. The benefits of this would be twofold. First, this would help to increase confidence in the identified ST-EPN-RELA and ST-EPN-YAP1 fusions. Second, this could help to identify rarer fusions among these subgroups that could also be targeted by ISPs.

Experimental validation of the peptides generated by the platform should also be conducted to confirm its predictive capability (Figure 2B). This would address the uncertainty posed by using machine learning algorithms and their predictions. To confirm the binding predictions by NetMHCpan-4.1 and NetMHCIIpan-4.0, the T-cell mediated immune response of each of the generated peptides should be tested. This can be done by quantifying the interferon gamma (IFN- γ) which is a cytokine secreted by activated CD4⁺ and CD8⁺ T cells [46]. This response can be measured using an enzyme-linked immunospot (ELISpot) assay [47]. The peptides that are confirmed to be immune stimulating should then go through animal and human clinical trials to determine their safety as a cancer treatment.

Finally, more work can be done to make the discovery platform more robust and automated. The machine learning algorithms allow for the prediction of epitope binding to specific MHC molecules, instead of supertypes. It would be interesting to update the platform

such that it can leverage a patient's own transcriptomic normal and tumor data to generate personalized immune therapies.

Conclusion

This study presents a novel discovery platform designed to harness transcriptomic data for the identification of fusion-generated neoantigens, to predict peptides that can initiate an anti-tumor immune response by targeting said neoantigens. Through the utilization machine learning, the platform factors in MHC binding and proteasomal cleavage predictions to maximize MHC supertype coverage. Application of this platform to two ependymoma subclasses prevalent in the pediatric patients successfully identified oncogenic fusions and generated 21 peptides predicted to elicit an anti-tumor immune response. Notably, the incorporation of cleavable KK and AAY linkers, alongside grouping binding epitopes into cleavage groups, introduces a novel approach to maximize the MHC allele coverage of a single peptide. This discovery platform emerges as a promising tool for advancing the development of novel cancer immunotherapies and personalized cancer therapy.

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Appendices

Appendix 1. Tables

Table 1. List of Ependymoma-patient transcriptomic data samples and meta data obtained from the SRA.

Run	Age at Diagnosis (Years)	Brain Location	Sex	Histological Grade	Molecular Subgroup	Primary/ Relapse
SRR15305023	1.3	Supratentorial	Male	III	ST-EPN-RELA	Relapse
SRR15305024	6.7	Supratentorial	Male	II	ST-EPN-RELA	Primary
SRR15305025	13.7	Supratentorial	Female	II	ST-EPN-RELA	Primary
SRR15305026	2.0	Supratentorial	Female	III	ST-EPN-YAP1	Relapse
SRR15305027	0.5	Supratentorial	Female	III	ST-EPN-YAP1	Primary
SRR15305028	1.0	Posterior Fossa	Male	II	PF-EPN-A	Relapse
SRR15305029	2.9	Posterior Fossa	Female	II	PF-EPN-A	Primary
SRR15305030	3.3	Posterior Fossa	Female	II	PF-EPN-A	Primary
SRR15305031	1.3	Posterior Fossa	Female	II	PF-EPN-A	Primary
SRR15305032	0.5	Posterior Fossa	Male	III	PF-EPN-A	Primary

Appendix 2. Figures

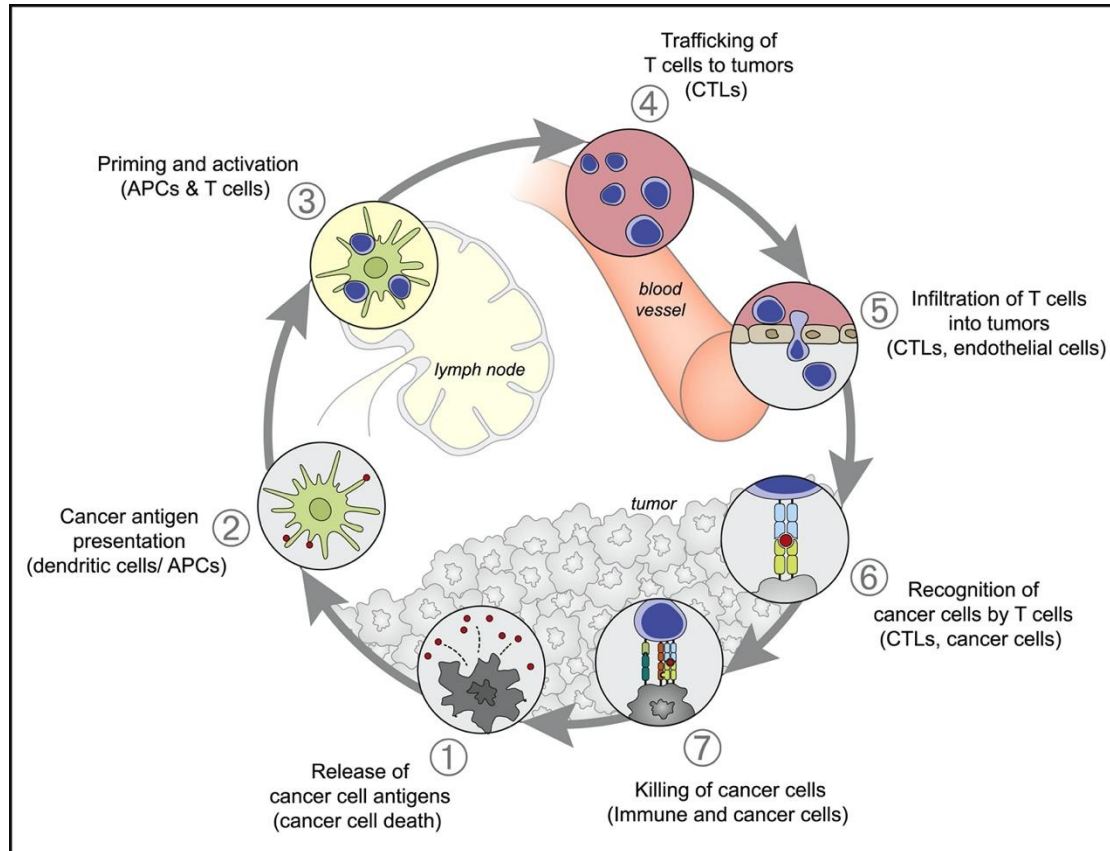


Figure 1. Pictorial Abstract of the Cancer Immunity Cycle, adapted from Chen and Mellman [9].

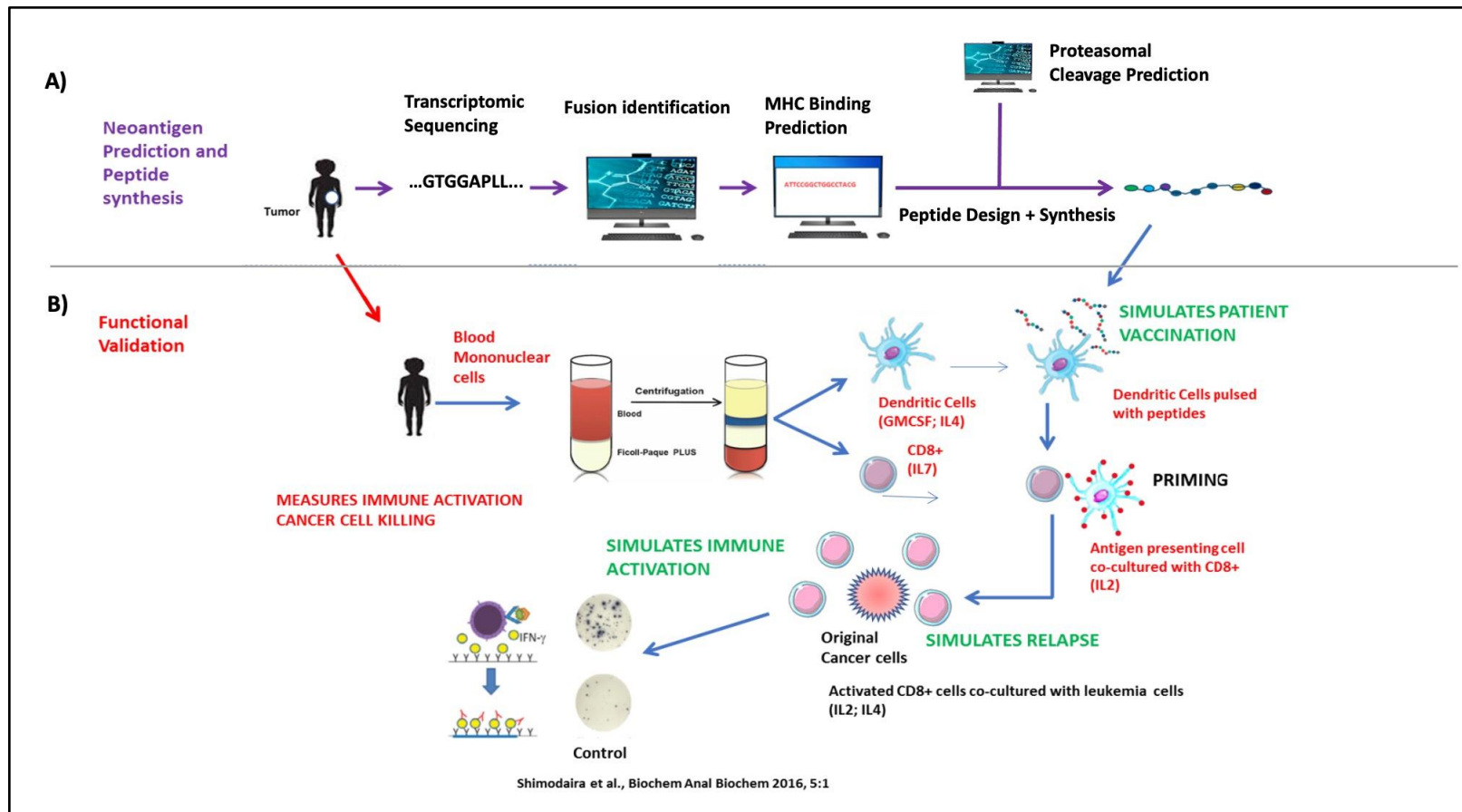


Figure 2. Diagram of ISP discovery and testing process. A) Bioinformatics discovery platform that was designed and used in this study. B) Future steps for validating the efficacy of the ISPs generated by the bioinformatics discovery platform.

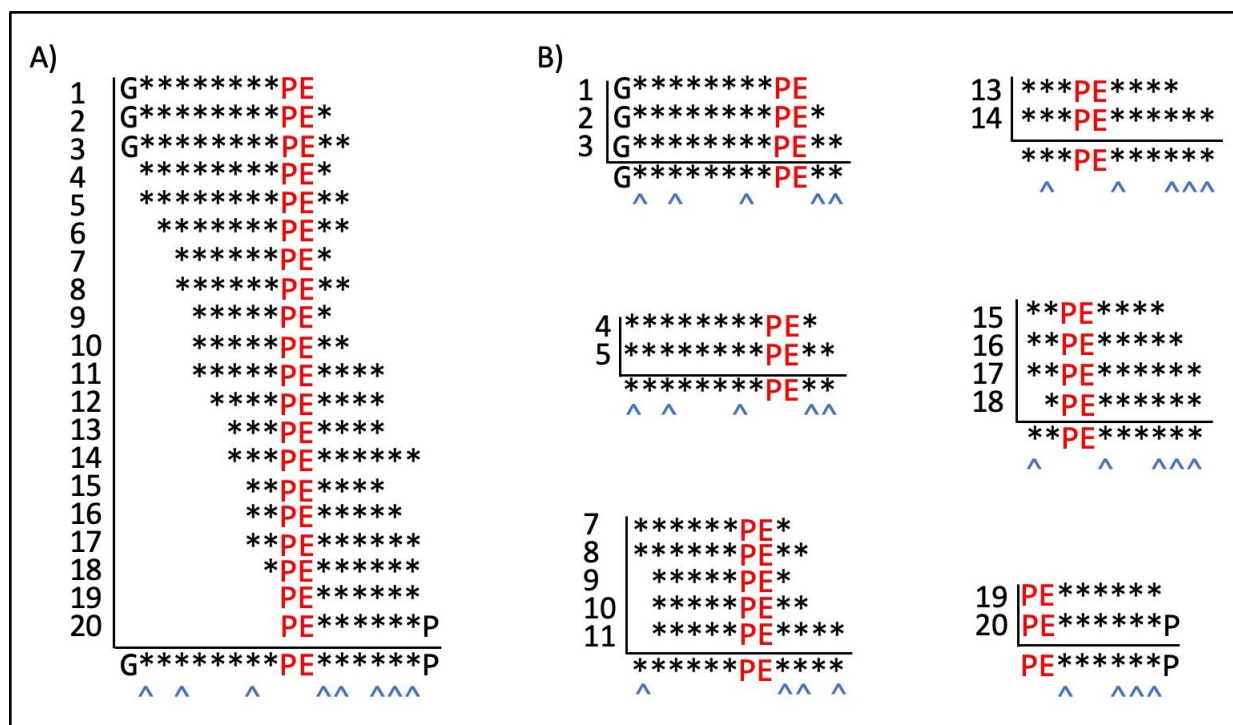


Figure 3. Predicted ST-EPN-RELA top binding epitopes across alleles with associated proteasomal cleavage sites. The proteasome is predicted to cleave the bond towards C-terminus of the indicated amino acid residue. A) Depiction of all 20 predicted ST-EPN-RELA binding epitopes with consensus peptide represented at the bottom. B) All 6 ST-EPN-RELA cleavage group peptides; the peptide at the bottom is expected to get cleaved into each of its above substituents.

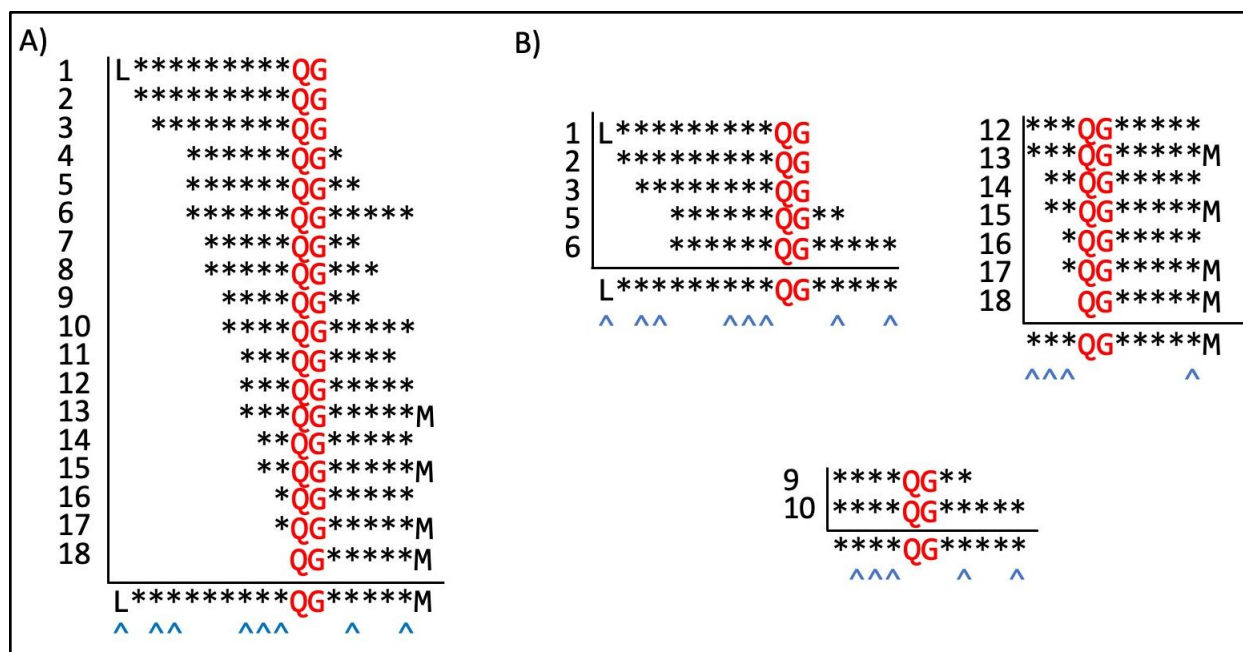


Figure 4. Predicted ST-EPN-YAP1 top binding epitopes across alleles with associated proteasomal cleavage sites. The proteasome is predicted to cleave the bond towards the C-terminus of the indicated amino acid residue. A) Depiction of all 18 predicted ST-EPN-YAP1 binding epitopes with consensus peptide represented at the bottom. B) All 3 ST-EPN-YAP1 cleavage group peptides; the peptide at the bottom is expected to get cleaved into each of its above substituents.

Appendix 3. Code

The python script to process the raw output data of netMHCpan and netMHCIIpan to retrieve the top predicted epitopes can be found at the following link: <https://github.com/SamLokanc/ISP>